

OPERATOR

off-board

Web GUI
joint RMS / peaks / thermal history

RViz2
robot + 2D / 3D maps

ros2 CLI / tools
pose, joint_test, param

rosbag2 / diagnostics
recording and replay

WebSocket JSON

topics + TF

services

ONBOARD COMPUTER

ROS 2; Humble local validation

robodog_web
aiohttp + WebSocket

cmd / telemetry

robodog_control
400 Hz loop, safety,
gait, balance, estimator

robot_state

robodog_perception
RGB-D, PointCloud2

render

robodog_description
URDF/Xacro, TF

read() / write()
registered RGB-D

RTAB-Map + OctoMap
default for supported MuJoCo;
map -> odom

robodog_hardware
JointBackend + RS06 CAN codec

robodog_sim
MuJoCo model + worlds

robodog_msgs
interface contract

UVC

not connected

HARDWARE / SOFTWARE BOUNDARY

JointBackend + CameraBackend

motion frames

PHYSICAL ROBOT

19.72 kg, 12 DOF

CAN bus 0
1 Mbit/s, front legs
6 x RS06, 72% load

CAN bus 1
1 Mbit/s, rear legs
6 x RS06, 72% load

NUWA HP60C
USB3 RGB; depth unavailable

IMU
integration unavailable

44.4 V battery
2 x 6S series, 2.50 kg

Physical and computational stack. Everything above the red boundary shares commands and telemetry; hardware travel is gated.